1️⃣ Competency 6: Identifying mathematical aspects of a real-world situation (Highest relevance)

Definition:

> Identify and describe the mathematical aspects of a real-world problem... formulate

🔍 How it appears in the task

The problem (page 1) provides:

- 4 rotations per minute
- Maximum of 2 people per sector
- 3 sectors in total
- Time duration: 30 minutes revolving-door-qs3

📌 Core insight

Students must extract:

- **Rate** (rotations per minute)
- **Capacity** (people per sector)
- **Structure** (3 sectors)
- **Time** (30 minutes)

👉 This reflects:

> "Extracting mathematical variables and relationships from a real-world context"

✔️ This is the most central *formulate* competency in this task

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2️⃣ Competency 2: Identifying key variables

Definition:

> Identify the key variables in a model formulate

🔍 Key variables

- Rotation rate: 4 rotations/min
- Capacity per rotation: 3 × 2 = 6 people
- Time: 30 minutes

📌 Key point

Students must recognize:

- One full rotation corresponds to one **cycle of people entering**

👉 Variable identification is essential for modeling

3️⃣ Competency 4: Understanding and constructing a model

Definition:

> Read, decode… to create a model of the situation formulate

🔍 How it appears in the task

Students construct a model:

- 1 rotation ⇒ 6 people
- 4 rotations per minute ⇒ 24 people per minute
- Over 30 minutes ⇒ total number of people

📌 Core idea

The real-world process is modeled as:

> **"rate × time = total quantity"**

👉 This is a classic process modeling task

II. Highly Relevant Competencies

4️⃣ Competency 5: Recognising mathematical structure

Definition:

> Recognise mathematical structure… formulate

🔍 How it appears in the task

The structure is:

- A **linear accumulation model**
- Core relationship:
  - Total people = (people per rotation) × (number of rotations)

📌 Mathematical structure

- Multiplicative structure
- Proportional reasoning

5️⃣ Competency 7: Simplifying and decomposing the problem

Definition:

> Simplify or decompose a situation... formulate

🔍 How it appears in the task

The continuous process is broken down into:

1. One rotation
2. One minute
3. Total time

👉 A hierarchical decomposition of the process

6️⃣ Competency 9: Translating into standard mathematical representation

Definition:

> Translate a problem into a standard mathematical representation formulate

🔍 How it appears in the task

The situation can be expressed as:

- Total = 4 × 30 × (3 × 2)

1️⃣ 能力6：识别现实问题中的数学要素（最高相关）

对应条目：

> Identify and describe the mathematical aspects of a real-world problem... formulate

🔍 题目体现

题目提供信息（第1页）：

- 每分钟旋转 4 圈
- 每个扇区最多容纳 2 人
- 一共有 3 个扇区
- 时间为 30 分钟

📌 本质分析

学生需要从情境中提取：

- **速率（rotations per minute）**
- **容量（people per sector）**
- **结构（3 sectors）**
- **时间（30 minutes）**

👉 属于：

> "从现实情境中抽取数学变量与关系"

✔️ 是本题最核心的 formulate 能力

2️⃣ 能力2：识别关键变量

对应条目：

> Identify the key variables in a model formulate

🔍 关键变量

- 转速：4 rotations/min
- 单次容量：3 × 2 = 6 人
- 时间：30 min

📌 关键点

学生必须明确：

- "一次完整旋转"对应一次"人员通过过程"

👉 变量识别是建模前提

3️⃣ 能力4：理解并构建问题模型

对应条目：

> Read, decode… to create a model of the situation formulate

🔍 题目体现

学生需构建模型：

- 每转一圈 ⇒ 6 人进入
- 每分钟 4 圈 ⇒ 每分钟 24 人
- 30 分钟 ⇒ 总人数

📌 本质

将现实过程建模为：

> **“速率 × 时间 = 总量”模型**

👉 属于典型过程建模

二、高相关能力

4️⃣ 能力5：识别数学结构

对应条目：

> Recognise mathematical structure… formulate

🔍 题目体现

结构为：

- **线性累积模型**
- 本质关系：
  - 总人数 =（每转人数）×（转数）

📌 数学结构

- 乘法结构
- 比例关系

5️⃣ 能力7：问题简化与分解

对应条目：

> Simplify or decompose a situation... formulate

🔍 题目体现

复杂过程（连续进人）被分解为：

1. 单次旋转
2. 每分钟
3. 总时间

👉 分解为多个层级计算

6️⃣ 能力9：转化为标准数学表示

对应条目：

> Translate a problem into a standard mathematical representation formulate

🔍 体现

可转化为：

- Total = 4 × 30 × (3 × 2)